

SCHOOL OF ELECTRONICS ENGINEERING

Fall Semester 2024-2025

BECE406E – FPGA Based System Design Lab

REG NO : 23BEC0358

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SLOT : L27 + L28

TASK 4 – Static Timing Analysis

Aim:

i) To design and implement a digital clock that displays time in the format Hours: Minutes: Seconds (HH:MM:SS) using the clock divider concept. The output is displayed using a seven segment display. Verify the functionality through simulation and implement the design on FPGA. Attach screenshots of code and FPGA board output.

ii) To design and implement a synchronous FIFO using Verilog in Quartus and verify its functionality using simulation. Perform static timing analysis and power analysis, and check for setup and hold time violations. Identify the critical path of the design. Attach the RTL view of FIFO and screenshots of code and outputs.

Software Details:

Functional Simulation: **Modelsim**

Synthesis & Implementation: **Intel Quartus Prime 23.1**

Hardware Details:

Family: Cyclone IV E

Package: FPGA

Pin count: 780

Core speed grade – 7

Device -EP4CE115F29C7

QUESTION 1)

i) Display the time in the format Hr: Min: Sec using clock divider concept and display the output using seven segment display. Attach screenshots of code and FPGA board implementation.

RTL Code for Digital Clock using Clock Divider and Seven Segment Display :

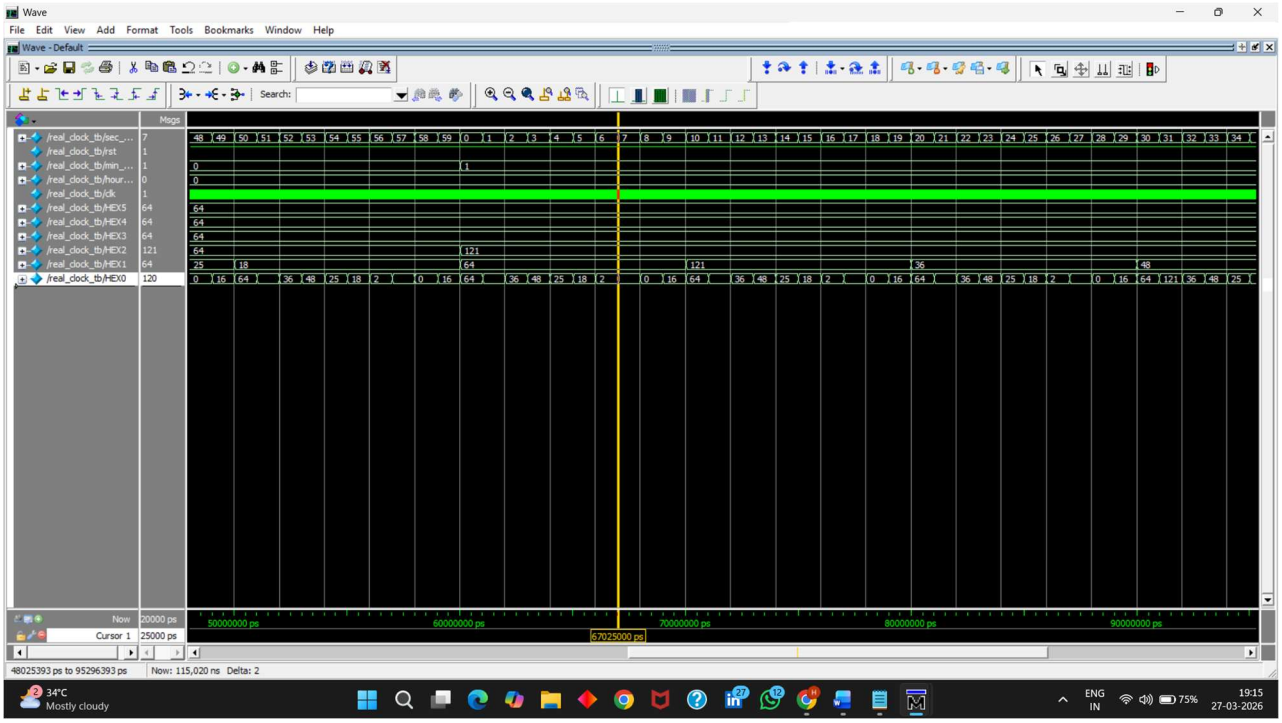
```
module real_clock( input clk,input rst,output [6:0] HEX0, output [6:0] HEX1,
   output [6:0] HEX2, output [6:0] HEX3, output [6:0] HEX4,  output [6:0] HEX5,
   output [5:0] sec_out,output [5:0] min_out, output [4:0] hour_out);
reg [25:0] count;reg tick_1hz;
always @(posedge clk or negedge rst) begin
   if(!rst) begin
      count <= 0;
      tick_1hz <= 0;
   end
   else if(count == 49_999_999) begin   // fast simulation
count <= 0;
      tick_1hz <= 1;
   end
   else begin
      count <= count + 1;
      tick_1hz <= 0;
   end
```

```
end
reg [5:0] sec;
reg [5:0] min;
reg [4:0] hour;
always @(posedge clk or negedge rst) begin
   if(!rst) begin
      sec <= 0;
      min <= 0;
      hour <= 0;
   end
   else if(tick_1hz) begin
      if(sec == 59) begin
         sec <= 0;
         if(min == 59) begin
            min <= 0;
            if(hour == 23)
               hour <= 0;
            else
               hour <= hour + 1;
         end
         else
            min <= min + 1;
end
      else
         sec <= sec + 1;
   end
end
assign sec_out  = sec;
```

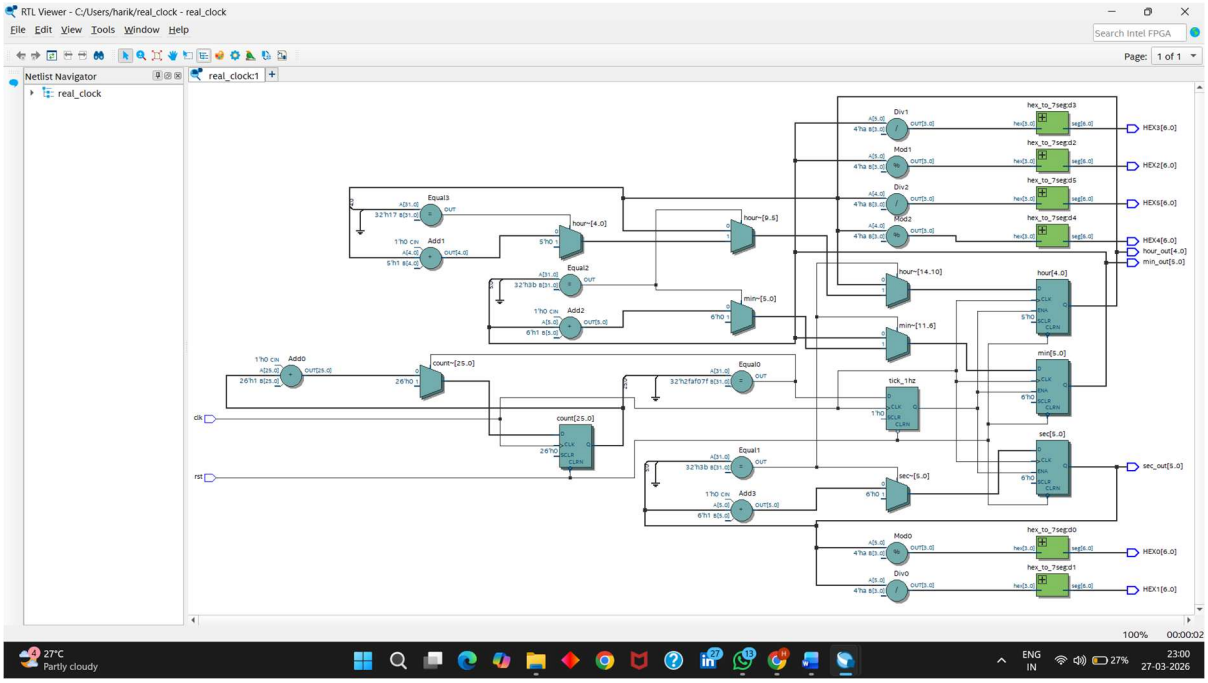
```
assign min_out  = min;assign hour_out = hour;
wire [3:0] sec_l  = sec % 10;
wire [3:0] sec_h  = sec / 10;
wire [3:0] min_l  = min % 10;
wire [3:0] min_h  = min / 10;
wire [3:0] hour_l = hour % 10;
wire [3:0] hour_h = hour / 10;
hex_to_7seg d0(sec_l, HEX0);
hex_to_7seg d1(sec_h, HEX1);
hex_to_7seg d2(min_l, HEX2);
hex_to_7seg d3(min_h, HEX3);
hex_to_7seg d4(hour_l, HEX4);
hex_to_7seg d5(hour_h, HEX5);
endmodule
module hex_to_7seg( input [3:0] hex,output reg [6:0] seg // active low);
always @(*) begin
   case(hex)
      4'h0: seg = 7'b1000000;
      4'h1: seg = 7'b1111001;
      4'h2: seg = 7'b0100100;
      4'h3: seg = 7'b0110000;
      4'h4: seg = 7'b0011001;
      4'h5: seg = 7'b0010010;
      4'h6: seg = 7'b0000010;
      4'h7: seg = 7'b1111000;
      4'h8: seg = 7'b0000000;
      4'h9: seg = 7'b0010000;
      default: seg = 7'b1111111;
```

```
    endcase
end
endmodule
`timescale 1ns/1ps
module real_clock_tb;
reg clk;reg rst;
wire [6:0] HEX0, HEX1, HEX2, HEX3, HEX4, HEX5;
wire [5:0] sec_out, min_out,wire [4:0] hour_out;
real_clock uut (
    .clk(clk),.rst(rst),
    .HEX0(HEX0),
    .HEX1(HEX1),
    .HEX2(HEX2),
    .HEX3(HEX3),
    .HEX4(HEX4),
 .HEX5(HEX5),
    .sec_out(sec_out),
    .min_out(min_out),
    .hour_out(hour_out));
always #5 clk = ~clk;
initial begin
    clk = 0;
    rst = 0;
    #20 rst = 1;
    #5000;
    $stop;
end
endmodule
```

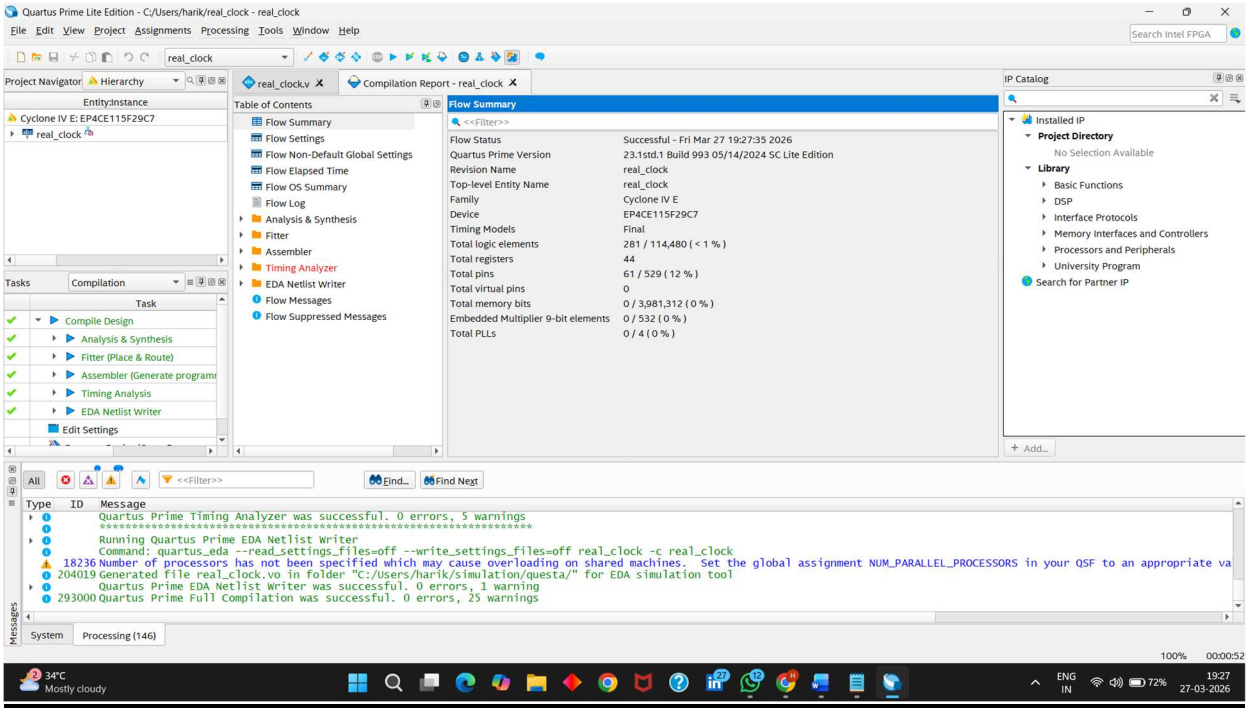
Functional Simulation of Digital Clock using ModelSim :



RTL View of Digital Clock Design :

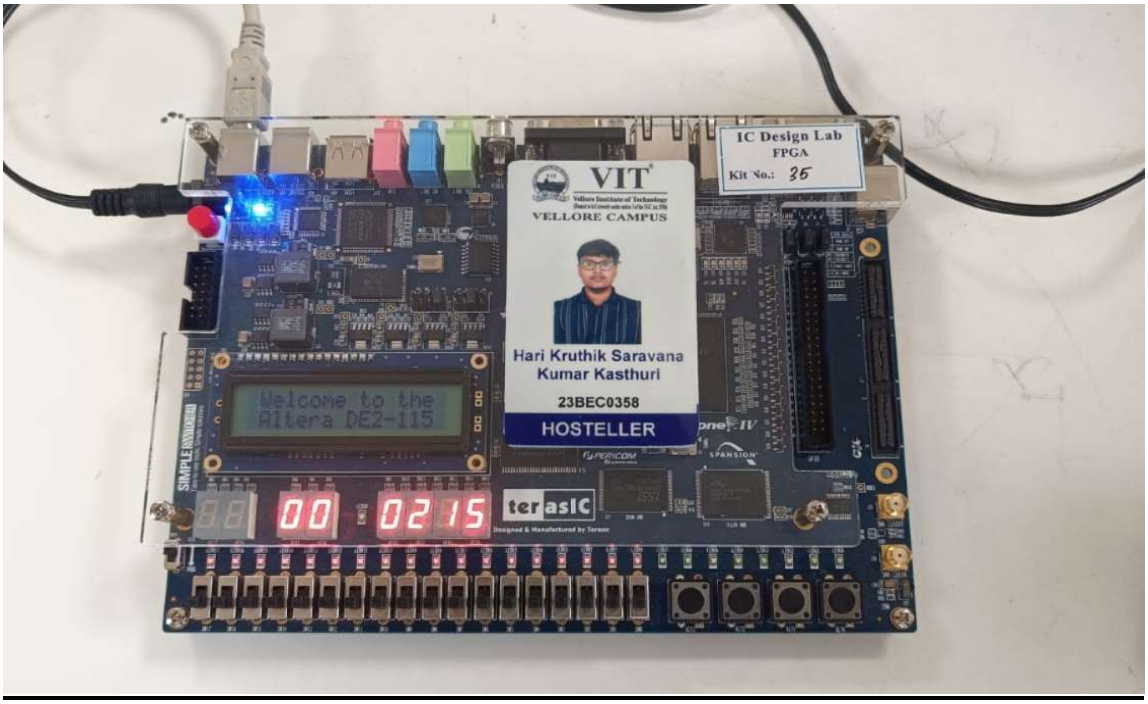


QUARTUS PRIME COMPILATION REPORT :

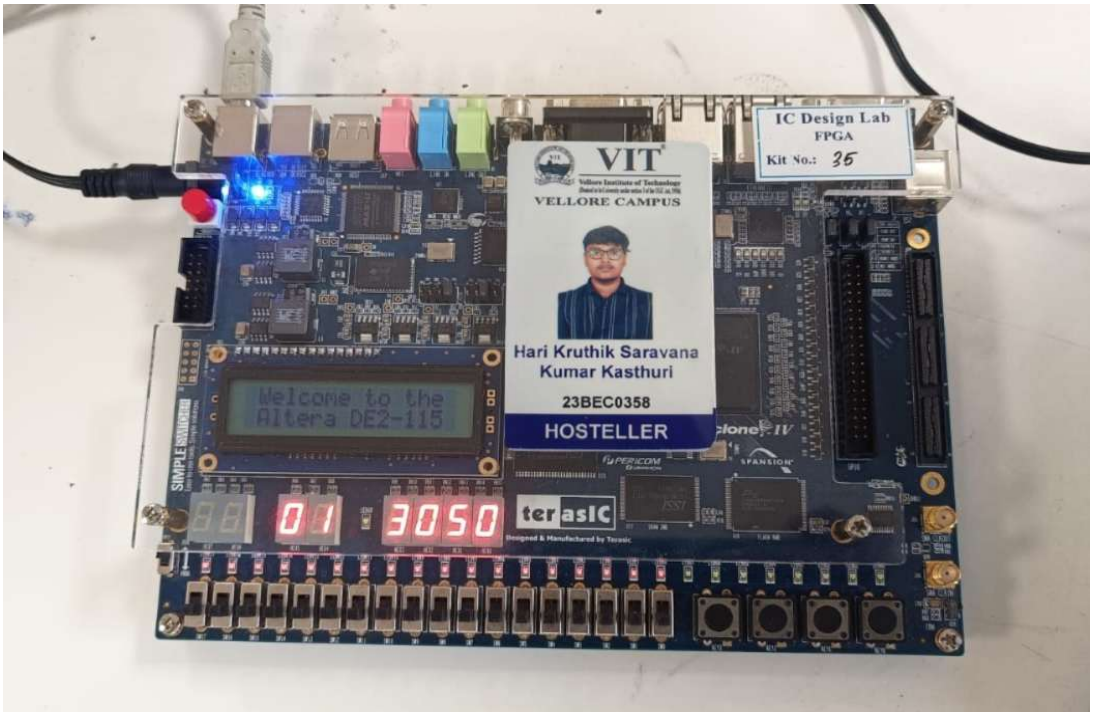


FPGA IMPLEMENTED PHOTOS :

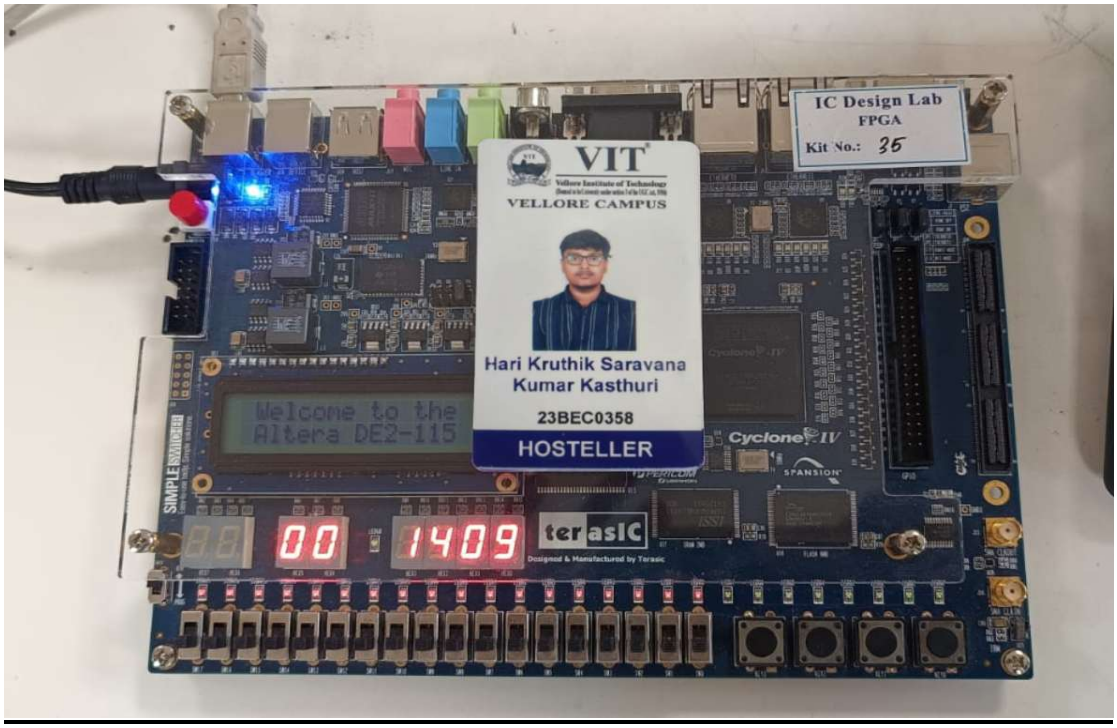
I)



II)


IC Design Lab
FPGA
Kit No.: 35
VIT
VELLORE CAMPUS
Hari Kruthik Saravana
Kumar Kasthuri
23BEC0358
HOSTELLER
Welcome to the
Altera DE2-115
01 3050
terasIC

III)


IC Design Lab
FPGA
Kit No.: 35
VIT
VELLORE CAMPUS
Hari Kruthik Saravana
Kumar Kasthuri
23BEC0358
HOSTELLER
Welcome to the
Altera DE2-115
00 1409
terasIC

QUESTION 2)

ii) Implement Synchronous FIFO using Quartus and verify the functionality. Verify the timing analysis and power analysis. Check for the set and hold violation and find the critical path. Attach the RTL view of FIFO and Screenshots of code and outputs.

RTL Code for Synchronous FIFO Design :

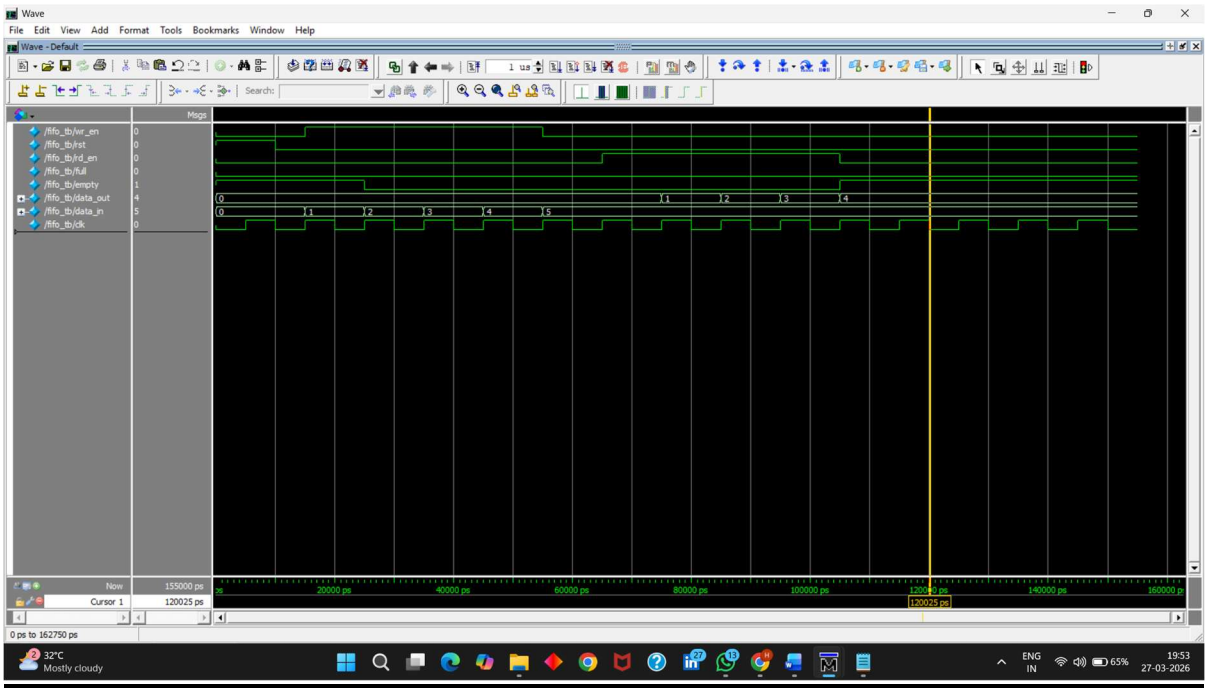
```
module fifo #(
   parameter WIDTH = 8,
   parameter DEPTH = 16)(
   input clk,
   input rst,
   input wr_en,
   input rd_en,
   input [WIDTH-1:0] data_in,
   output reg [WIDTH-1:0] data_out,
   output full,
   output empty);
reg [WIDTH-1:0] mem [0:DEPTH-1];
reg [$clog2(DEPTH)-1:0] wr_ptr, rd_ptr;
reg [$clog2(DEPTH):0] count;
always @(posedge clk or posedge rst) begin
   if (rst)
      wr_ptr <= 0;
   else if (wr_en && !full) begin
```

```
        mem[wr_ptr] <= data_in;
        wr_ptr <= wr_ptr + 1;
end
end
always @(posedge clk or posedge rst) begin
    if (rst) begin
        rd_ptr <= 0;
        data_out <= 0;
    end
    else if (rd_en && !empty) begin
        data_out <= mem[rd_ptr];
        rd_ptr <= rd_ptr + 1;
    end
end
always @(posedge clk or posedge rst) begin
    if (rst)
        count <= 0;
    else begin
        case ({wr_en && !full, rd_en && !empty})
            2'b10: count <= count + 1; // write
            2'b01: count <= count - 1; // read
            default: count <= count;
        endcase
    end
end
assign full  = (count == DEPTH);
assign empty = (count == 0);
endmodule
```

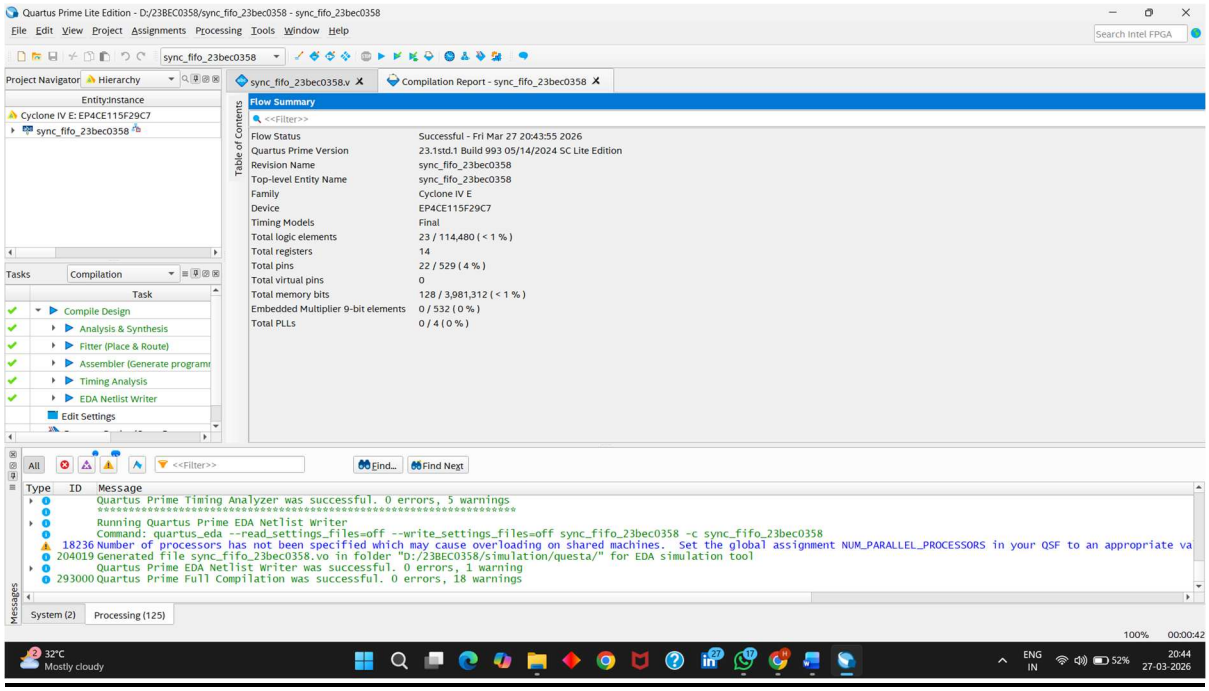
```
`timescale 1ns/1ps
module fifo_tb;
reg clk;
reg rst;
reg wr_en, rd_en;
reg [7:0] data_in;
wire [7:0] data_out;
wire full, empty;
fifo uut (
    .clk(clk),
    .rst(rst),
    .wr_en(wr_en),
    .rd_en(rd_en),
    .data_in(data_in),
    .data_out(data_out),
    .full(full),
    .empty(empty));
always #5 clk = ~clk;
initial begin
    clk = 0;
    rst = 1;
    wr_en = 0;
    rd_en = 0;
    data_in = 0;
    #10 rst = 0;
    repeat(5) begin
        @(posedge clk);
        wr_en = 1;
```

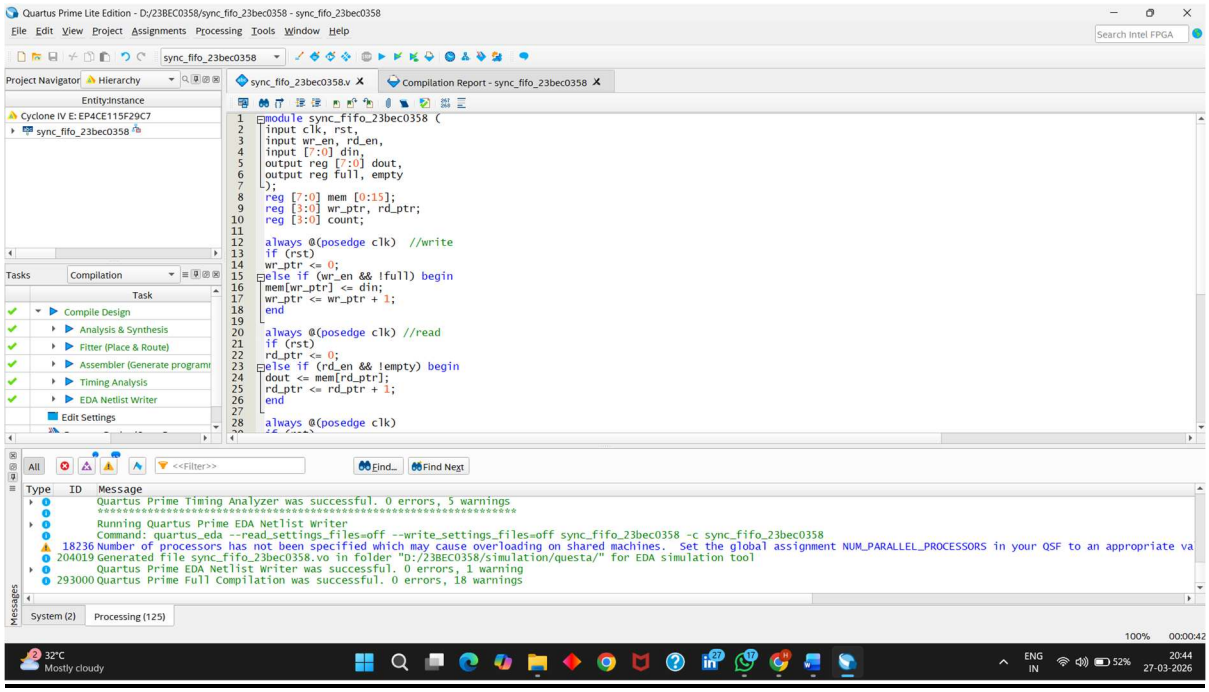
```
        data_in = data_in + 1;
    end
    wr_en = 0;
    repeat(5) begin
        @(posedge clk);
        rd_en = 1;
    end
    rd_en = 0;
    #50 $stop;
end
endmodule
```

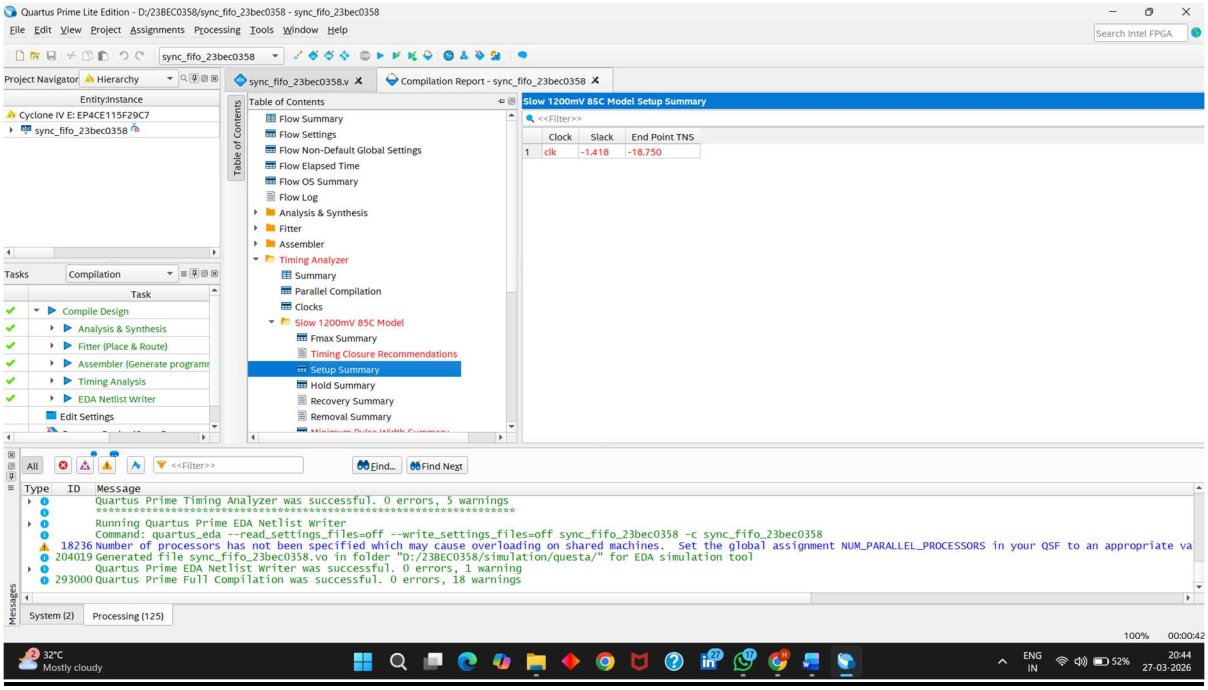
Functional Simulation of Synchronous FIFO using ModelSim :

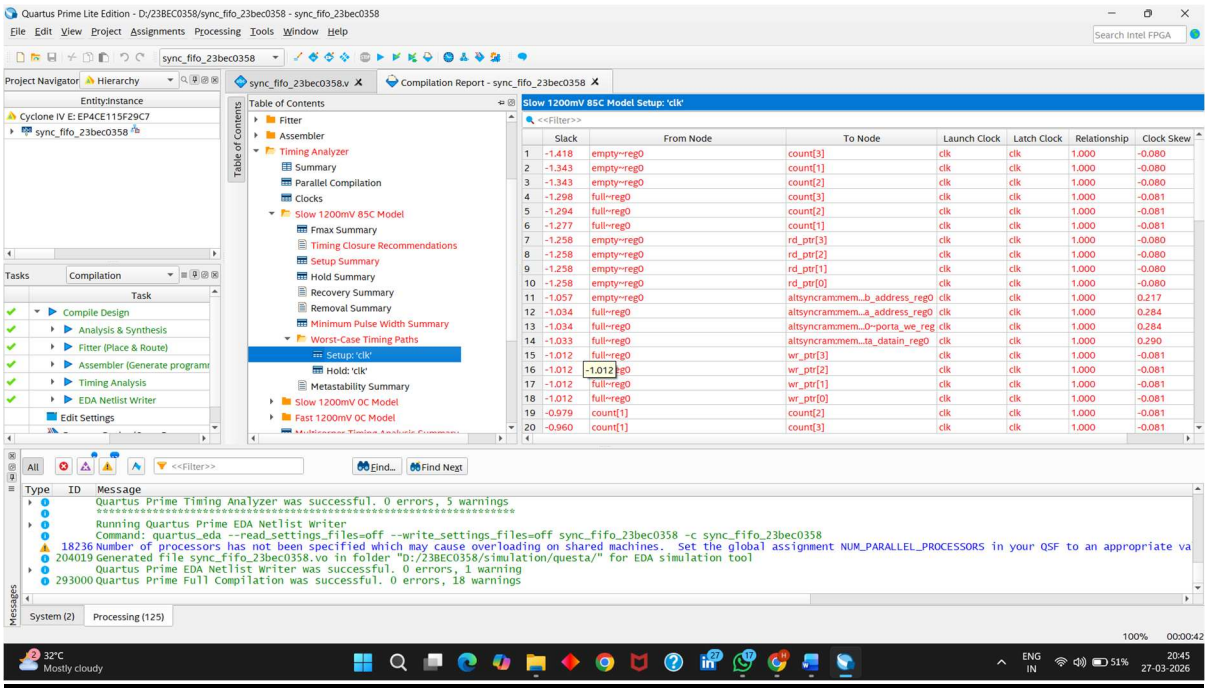


QUARTUS PRIME COMPILATION REPORT :[before Timing Analysis - sdc file]

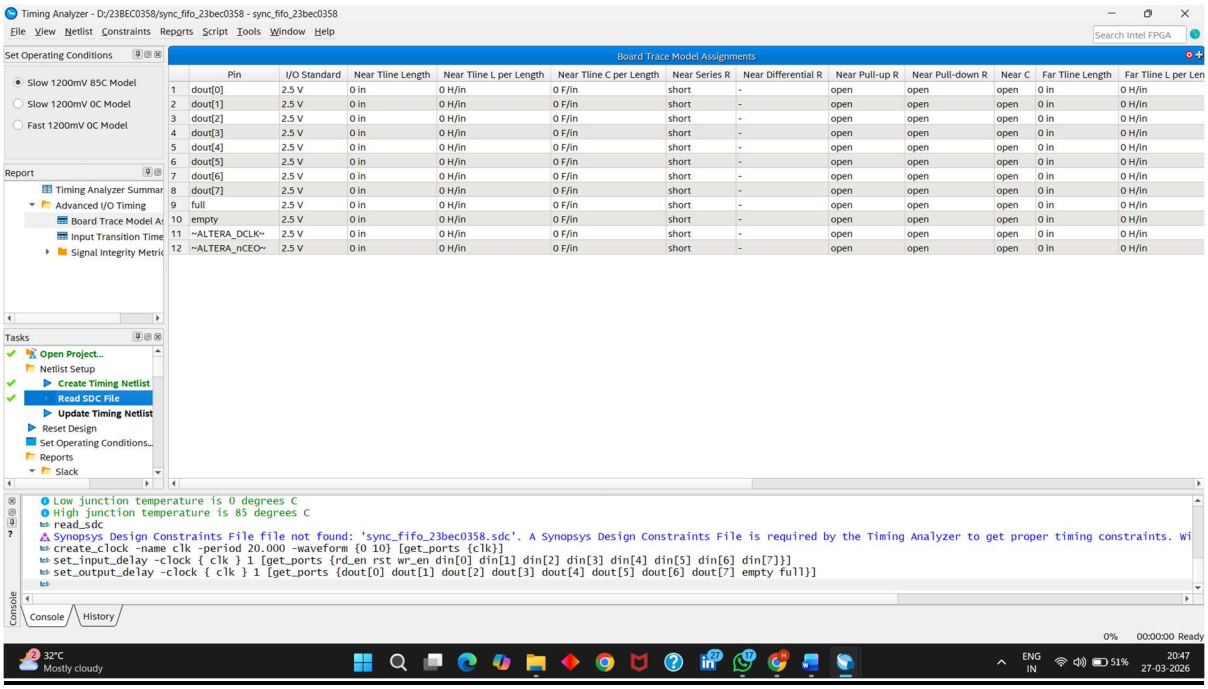


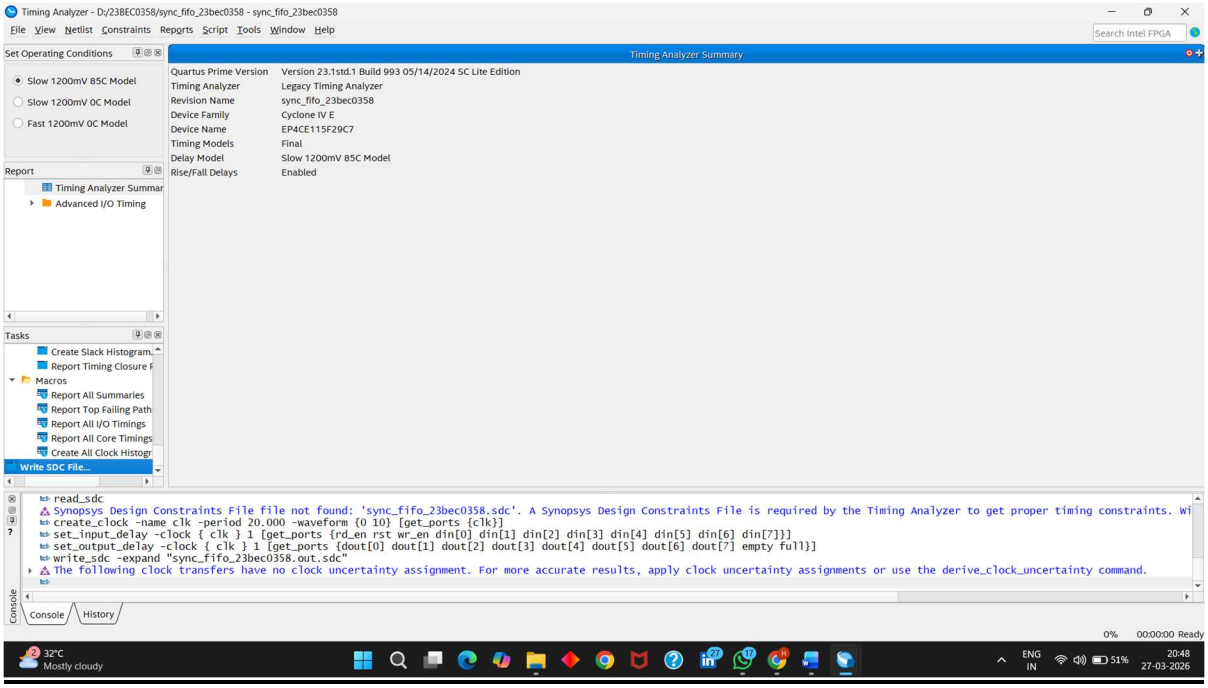



Quartus Prime Lite Edition - D:/23BEC0358/sync_fifo_23bec0358 - sync_fifo_23bec0358
File Edit View Project Assignments Processing Tools Window Help
Slow 1200mV 85C Model Setup Summary
Clock Slack End Point TNS
clk -1.418 -18.750
Quartus Prime Timing Analyzer was successful. 0 errors, 5 warnings
Running Quartus Prime EDA Netlist Writer
Command: quartus_eda --read_settings_files=off --write_settings_files=off sync_fifo_23bec0358 -c sync_fifo_23bec0358
18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your QSF to an appropriate va
204019 Generated file sync_fifo_23bec0358.vo in folder "D:/23BEC0358/simulation/questa/" for EDA simulation tool
Quartus Prime EDA Netlist Writer was successful. 0 errors, 1 warning
293000 Quartus Prime Full Compilation was successful. 0 errors, 18 warnings

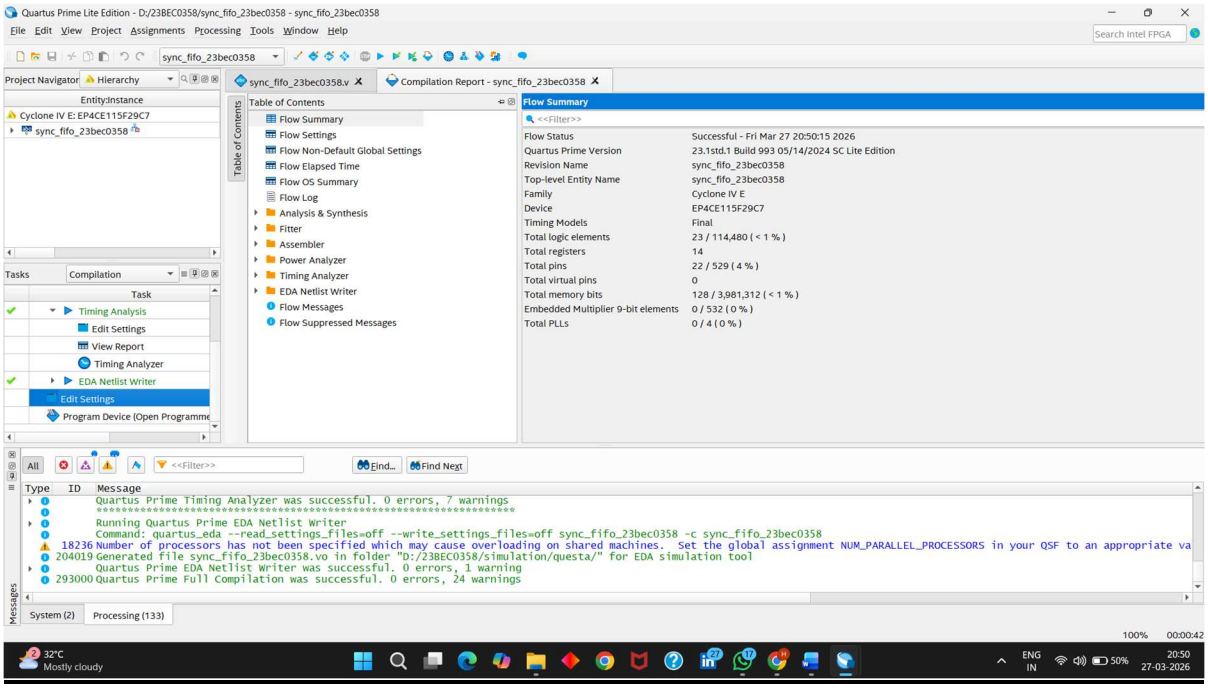

Quartus Prime Lite Edition - D:/23BEC0358/sync_fifo_23bec0358 - sync_fifo_23bec0358
Slow 1200mV 85C Model Setup: 'clk'
Slack From Node To Node Launch Clock Latch Clock Relationship Clock Skew
1 -1.418 empty~reg0 count[3] clk clk 1.000 -0.080
2 -1.343 empty~reg0 count[1] clk clk 1.000 -0.080
3 -1.343 empty~reg0 count[2] clk clk 1.000 -0.080
4 -1.298 full~reg0 count[3] clk clk 1.000 -0.081
5 -1.294 full~reg0 count[2] clk clk 1.000 -0.081
6 -1.277 full~reg0 count[1] clk clk 1.000 -0.081
7 -1.258 empty~reg0 rd_ptr[3] clk clk 1.000 -0.080
8 -1.258 empty~reg0 rd_ptr[2] clk clk 1.000 -0.080
9 -1.258 empty~reg0 rd_ptr[1] clk clk 1.000 -0.080
10 -1.258 empty~reg0 rd_ptr[0] clk clk 1.000 -0.080
11 -1.057 empty~reg0 altsyncram:mem...b_address_reg0 clk clk 1.000 0.217
12 -1.034 full~reg0 altsyncram:mem...a_address_reg0 clk clk 1.000 0.284
13 -1.034 full~reg0 altsyncram:mem...0~porta_we_reg clk clk 1.000 0.284
14 -1.033 full~reg0 altsyncram:mem...ta_datain_reg0 clk clk 1.000 0.290
15 -1.012 full~reg0 wr_ptr[3] clk clk 1.000 -0.081
16 -1.012 -1.012 wr_ptr[2] clk clk 1.000 -0.081
17 -1.012 full~reg0 wr_ptr[1] clk clk 1.000 -0.081
18 -1.012 full~reg0 wr_ptr[0] clk clk 1.000 -0.081
19 -0.979 count[1] count[2] clk clk 1.000 -0.081
20 -0.960 count[1] count[3] clk clk 1.000 -0.081
Quartus Prime Timing Analyzer was successful. 0 errors, 5 warnings
Running Quartus Prime EDA Netlist Writer
Command: quartus_eda --read_settings_files=off --write_settings_files=off sync_fifo_23bec0358 -c sync_fifo_23bec0358
18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your QSF to an appropriate va
204019 Generated file sync_fifo_23bec0358.vo in folder "D:/23BEC0358/simulation/questa/" for EDA simulation tool
Quartus Prime EDA Netlist Writer was successful. 0 errors, 1 warning
293000 Quartus Prime Full Compilation was successful. 0 errors, 18 warnings

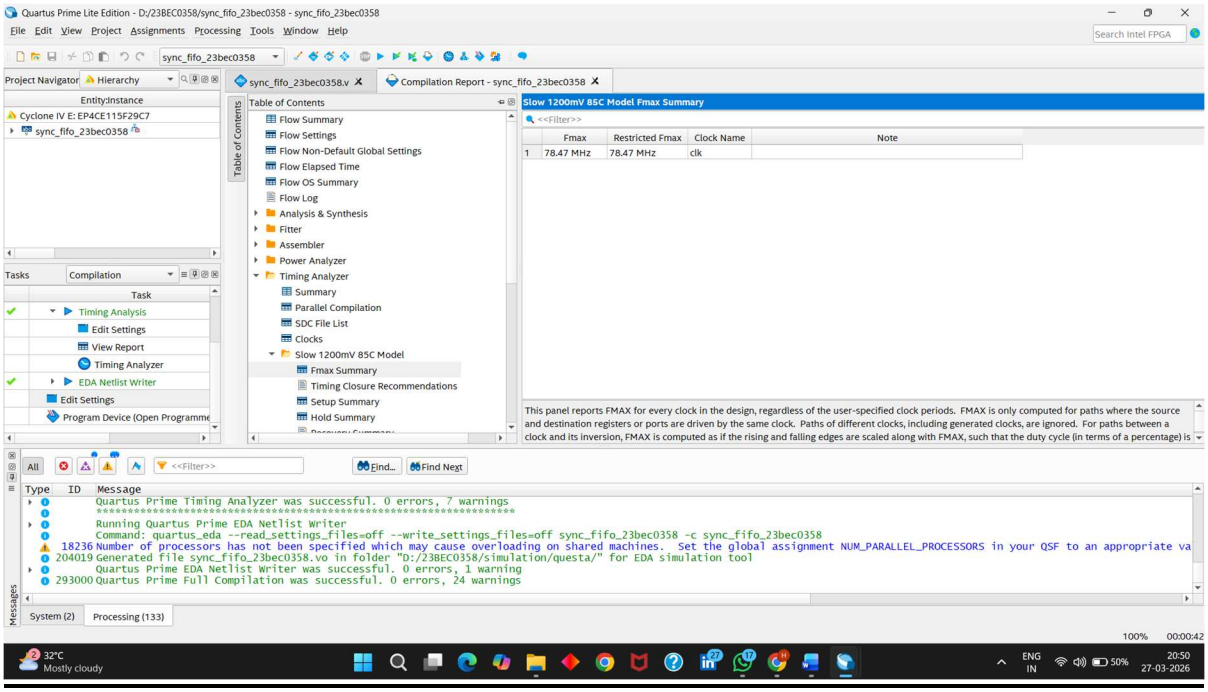
Static Timing Analysis of FIFO using Timing Analyzer :

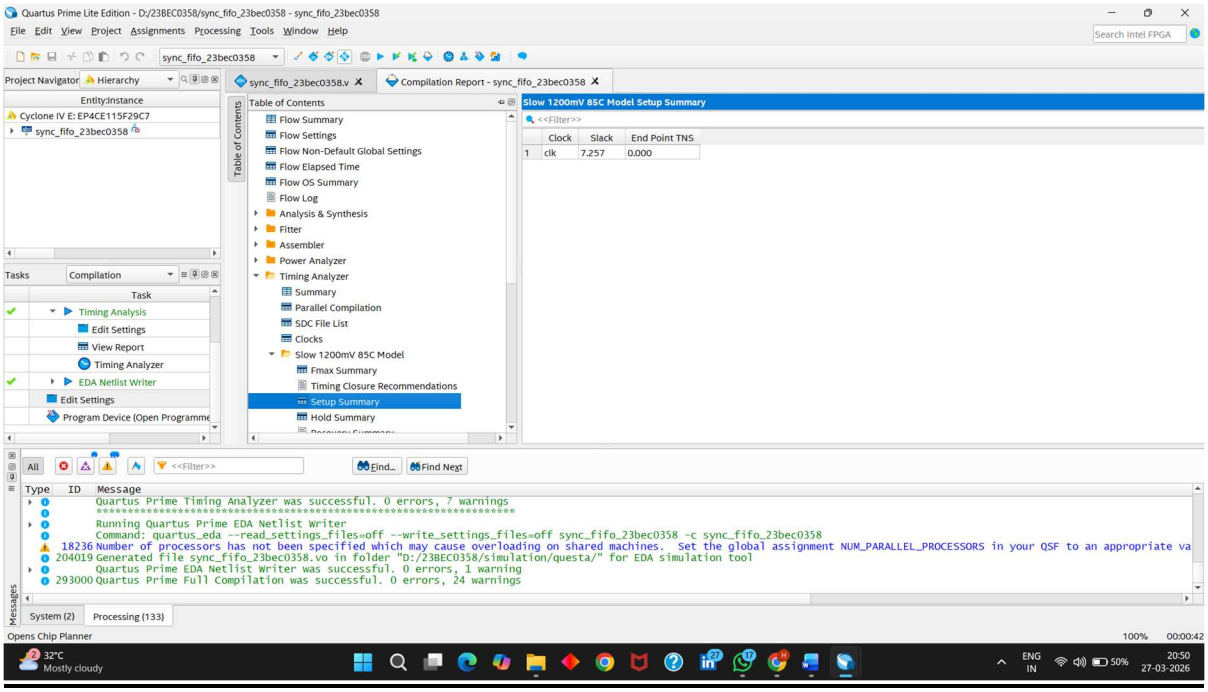


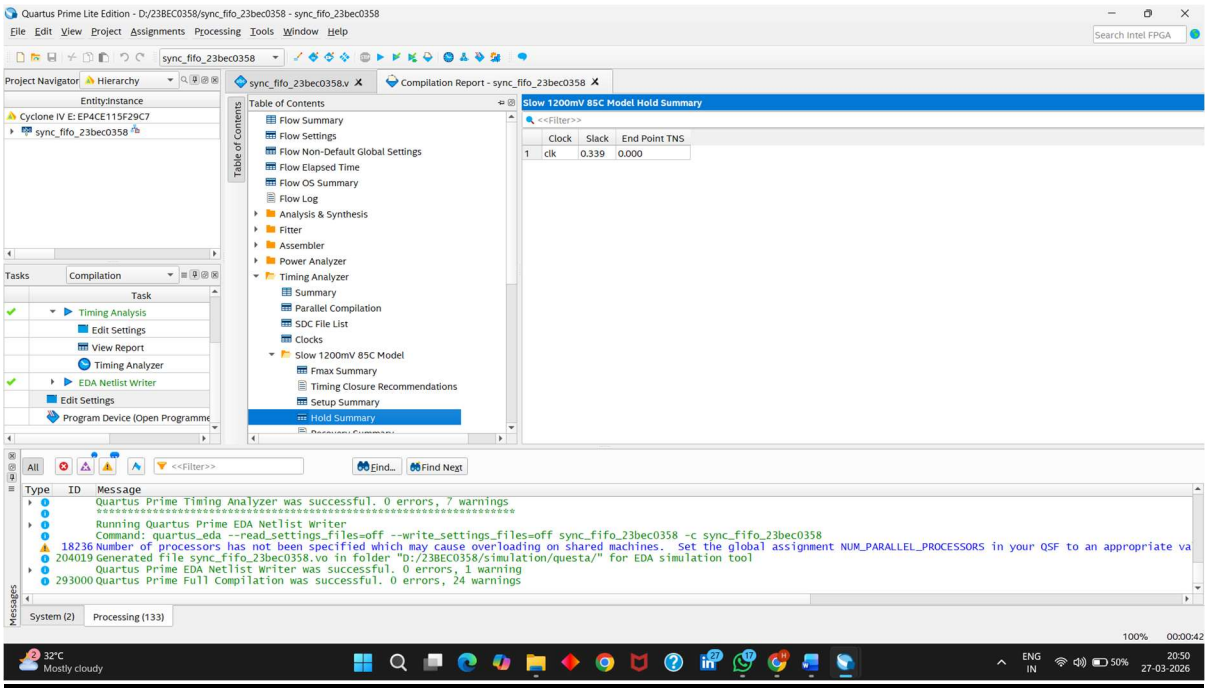


QUARTUS PRIME COMPILATION REPORT :[after Timing Analysis - sdc file]

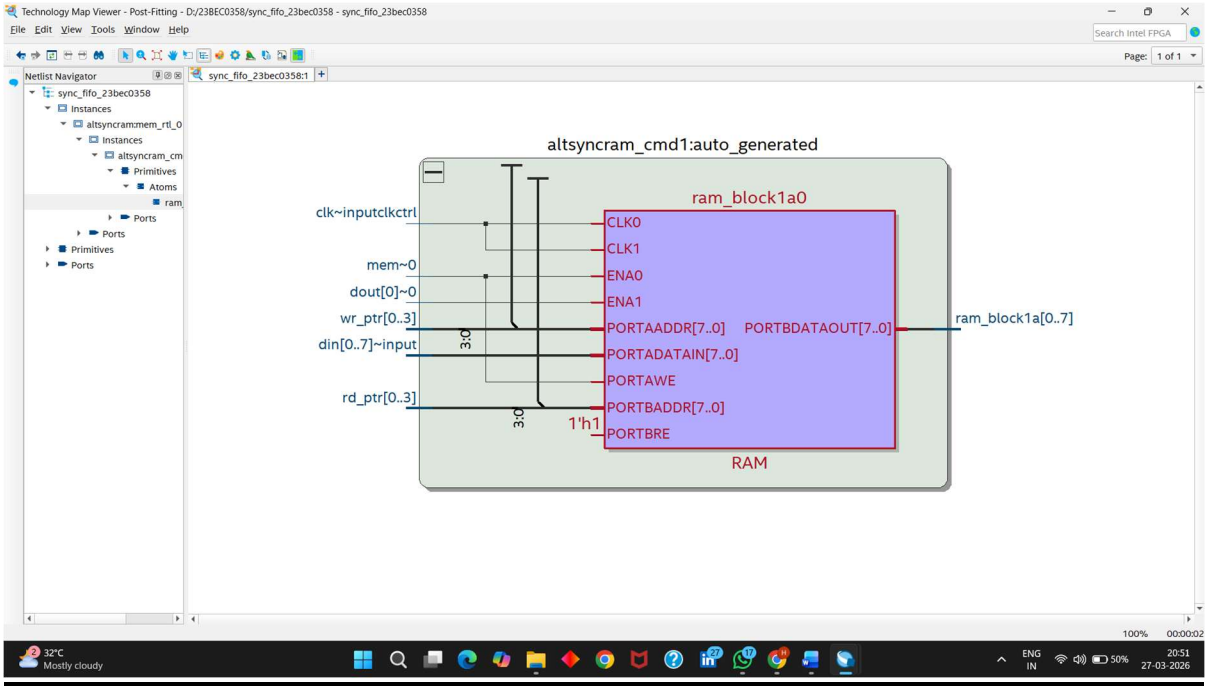




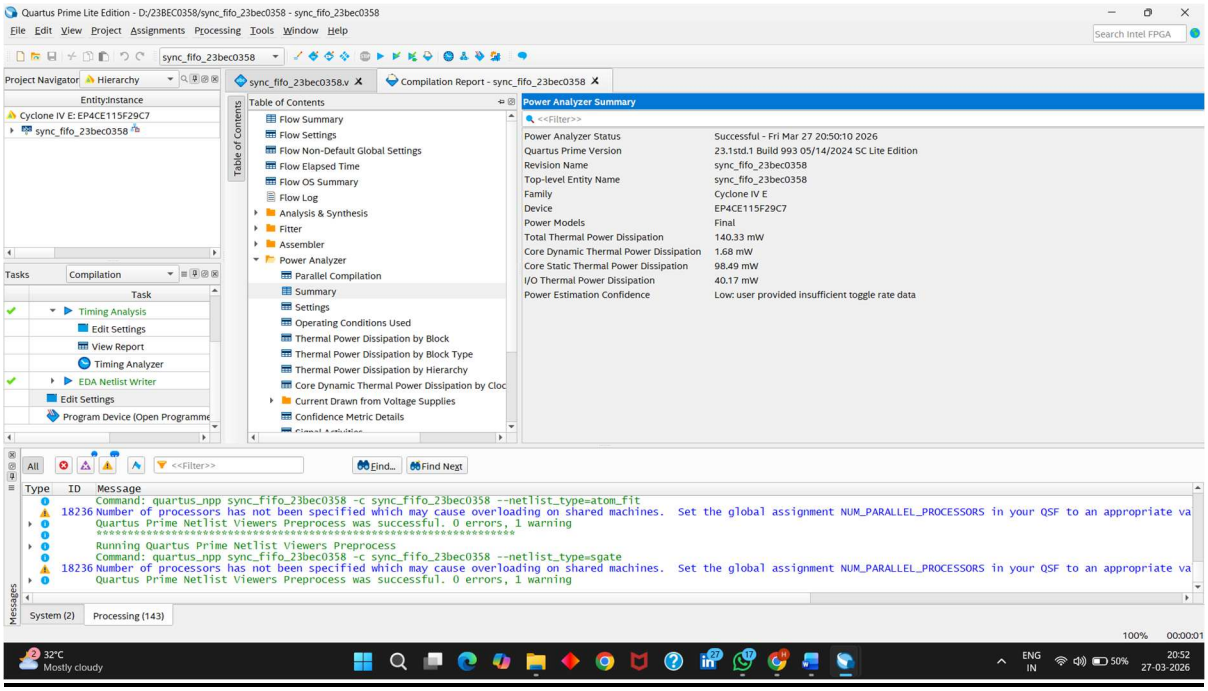

Quartus Prime Lite Edition - D:/23BEC0358/sync_fifo_23bec0358 - sync_fifo_23bec0358
File Edit View Project Assignments Processing Tools Window Help
Search Intel FPGA
Project Navigator Hierarchy
Entity:Instance
Cyclone IV E: EP4CE115F29C7
sync_fifo_23bec0358
Tasks Compilation
Task
Timing Analysis
Edit Settings
View Report
Timing Analyzer
EDA Netlist Writer
Edit Settings
Program Device (Open Programme
sync_fifo_23bec0358.v
Compilation Report - sync_fifo_23bec0358
Table of Contents
Flow Summary
Flow Settings
Flow Non-Default Global Settings
Flow Elapsed Time
Flow OS Summary
Flow Log
Analysis & Synthesis
Fitter
Assembler
Power Analyzer
Timing Analyzer
Summary
Parallel Compilation
SDC File List
Clocks
Slow 1200mV 85C Model
Fmax Summary
Timing Closure Recommendations
Setup Summary
Hold Summary
Slow 1200mV 85C Model Setup Summary
<<Filter>>
Clock Slack End Point TNS
1 clk 7.257 0.000
All
<<Filter>>
Find... Find Next
Type ID Message
Quartus Prime Timing Analyzer was successful. 0 errors, 7 warnings
Running Quartus Prime EDA Netlist Writer
Command: quartus_eda --read_settings_files=off --write_settings_files=off sync_fifo_23bec0358 -c sync_fifo_23bec0358
18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your QSF to an appropriate va
204019 Generated file sync_fifo_23bec0358.vo in folder "D:/23BEC0358/simulation/questa/" for EDA simulation tool
Quartus Prime EDA Netlist Writer was successful. 0 errors, 1 warning
293000 Quartus Prime Full Compilation was successful. 0 errors, 24 warnings
Messages
System (2) Processing (133)
Opens Chip Planner
100% 00:00:42
32°C Mostly cloudy
ENG IN 50% 20:50 27-03-2026


Quartus Prime Lite Edition - D:/23BEC0358/sync_fifo_23bec0358 - sync_fifo_23bec0358
File Edit View Project Assignments Processing Tools Window Help
Search Intel FPGA
Project Navigator Hierarchy
Entity:Instance
Cyclone IV E: EP4CE115F29C7
sync_fifo_23bec0358
Tasks Compilation
Task
Timing Analysis
Edit Settings
View Report
Timing Analyzer
EDA Netlist Writer
Edit Settings
Program Device (Open Programme
sync_fifo_23bec0358.v
Compilation Report - sync_fifo_23bec0358
Table of Contents
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Timing Analyzer
Summary
Parallel Compilation
SDC File List
Clocks
Slow 1200mV 85C Model
Fmax Summary
Timing Closure Recommendations
Setup Summary
Hold Summary
Slow 1200mV 85C Model Hold Summary
<<Filter>>
Clock Slack End Point TNS
1 clk 0.339 0.000
All
<<Filter>>
Find... Find Next
Type ID Message
Quartus Prime Timing Analyzer was successful. 0 errors, 7 warnings
Running Quartus Prime EDA Netlist Writer
Command: quartus_eda --read_settings_files=off --write_settings_files=off sync_fifo_23bec0358 -c sync_fifo_23bec0358
18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your QSF to an appropriate va
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Quartus Prime EDA Netlist Writer was successful. 0 errors, 1 warning
293000 Quartus Prime Full Compilation was successful. 0 errors, 24 warnings
Messages
System (2) Processing (133)
100% 00:00:42
32°C Mostly cloudy
ENG IN 50% 20:50 27-03-2026

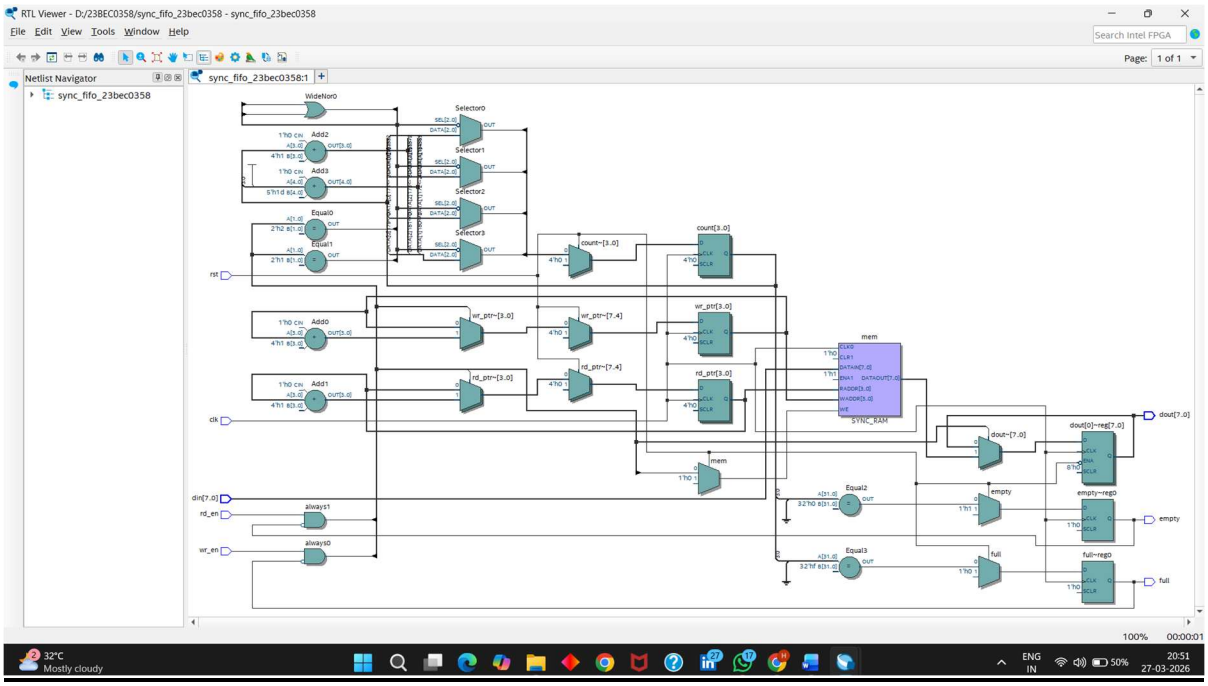
Critical Path Identification in FIFO Design :



Power Analysis of FIFO using Quartus Power Analyzer :



RTL View of FIFO Design :



INFERENCE :

The digital clock was designed using a clock divider and implemented to display time in the format of hours, minutes, and seconds on a seven segment display. The simulation results confirmed correct time progression with proper increment of seconds, minutes, and hours. The design functioned as expected and demonstrated reliable operation using synchronous logic.

The synchronous FIFO was implemented and verified through simulation, showing correct First-In-First-Out behavior with proper read and write operations. From the timing analysis, the initial results showed negative slack (setup slack ≈ -1.418 ns), indicating a timing violation. After optimization, the design achieved positive slack (setup slack ≈ 7.257 ns and hold slack ≈ 0.339 ns), confirming that the timing requirements are satisfied and the design operates reliably. The maximum operating frequency (Fmax) was observed to be approximately 78.47 MHz. Power analysis indicated a total thermal power dissipation of around 140 mW, showing that the design is efficient in terms of power consumption. The critical path was identified as the longest delay path affecting the system performance.